

Color Management in Virtual Production

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1 INTRODUCTION

Lux Machina presents their developments in color management for virtual production in this talk. The company has extensive experience in delivering innovative and immersive experiences for live events, film, and television, and their color management techniques have played a critical role in achieving those experiences. The talk begins by discussing the challenges faced in production today. It covers developments in both hardware, software, and workflow in pursuit to address them. These solutions enable them to achieve accurate and consistent color throughout the virtual production process, from pre-visualization to capture and into post. This talk will touch on a variety of topics: novel approaches to measurement, monitoring, modifications to Unreal Engine, calibration of lighting and displays for cameras, and developing specialized software tools for color management in virtual production.

Overall, Lux Machina's developments in color management demonstrate the company's commitment to delivering immersive and engaging experiences for their clients. The company's hardware and software tooling, approaches to production challenges, and visualization techniques are all designed to ensure accurate and consistent color, and the talk provides valuable insights into the company's process.

2 PRESENTATION BREAKDOWN

The next subsections provide topics that will be included in the talk.

2.1 Current challenges

ICVFX refers to a filmmaking technique that combines real-time computer graphics and live-action footage to create seamless and immersive environments for film, television, or other media productions. It involves the integration of virtual elements, such as virtual sets, characters, and visual effects, with live actors and physical sets in real-time during the production process. As virtual production is still a relatively new approach, it presents unique challenges that

have not been encountered before. Some of these challenges stem from the software used in virtual production, as the requirements and demands of filmmakers in virtual production sets couldn't have been predicted in advance. On the other hand, hardware-related challenges arise due to the fundamental functioning of devices. Display technology, originally optimized for human perception and final viewing environments, now needs to serve as a lighting instrument and be observed by the camera. These challenges call for ongoing advancements and adaptations in both software and hardware to cater to the specific needs and demands of virtual production. Solving these challenges often requires planning ahead, time and money investments and a good understanding of the science behind. Our talk will try and shed some light on possible approaches to solving these issues.

2.2 Assets, Filetypes, Color Spaces, Signal Paths, Monitoring

To address these challenges, the initial focus lies in establishing an efficient and effective pipeline for handling pixel data. This involves examining asset management, determining the most suitable file types, and selecting appropriate color spaces to maximize the transmission of information. Although creating entirely new color spaces may not be necessary, it is crucial to identify the most suitable options for our specific use case and each step of the process. Considering the intricacies of the production chain, alternative color spaces might offer better performance and fidelity. By carefully evaluating and selecting the appropriate color spaces, we can ensure that the pixel data flows smoothly and effectively throughout the virtual production pipeline, leading to improved results.

We will also look at how we can monitor all of the assets on sets and carefully convey to all teams the creative intent. Correctly monitoring assets on sets can usually present a challenge, especially since there usually is a mix of dynamic ranges, peak brightnesses and color gamuts. We can offer creatives different options in how they want to view the feeds in ways that suit them most. We can also offer creatives the ability to pre-visualize most assets through the creative LUT that will be applied on-set so that they can view exactly how their assets will be perceived once in production.

2.3 Validation and calibration

As previously discussed, the existing display hardware was not initially intended for observation by cameras. This poses a distinct challenge, as it requires us to meticulously calibrate our displays and observers to ensure that the assets created by artists remain unaltered as they pass through the imaging chain. Traditionally, display calibrations can be a time-consuming and labor-intensive process. However, we will introduce novel methods, through which we aim to optimize display calibrations, while minimizing their impact on

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overall production time and resources. Our novel calibration approaches often involve the display of a series of test patterns, which are then recorded. However, this conventional method requires significant time and expertise. To address this, we have developed tools that allow for the rapid display of patches, eliminating the need for additional software processing. These tools enable us to achieve efficient Camera-LED color calibrations without relying solely on color scientists. We will demonstrate how operators, who may not necessarily have extensive color science knowledge, can effectively execute these calibrations in an efficient manner. Since

we also have control over traditional lighting instruments, we will look at how we can make sure they are also correctly conveying color information when viewed by a camera. Oftentimes we can use calibrated modes of lights and characterize how the camera perceives them in similar ways to how we treat the LED displays.

2.4 Looking ahead

Lastly, we want to look toward the future and talk about emerging technologies that will help alleviate some of the challenges we are presented with.